

AI-POWERED INTELLIGENT PROCESS OPTIMIZATION

GRADE SENSE

Predict quality. Explain risk. Recommend action.

An AI-assisted manufacturing intelligence platform that predicts product quality in real time and recommends safe process adjustments before defects occur.

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Honeywell Hackathon · Industrial AI / Smart Manufacturing

PREDICT



EXPLAIN



RECOMMEND

HUMAN-IN-THE-LOOP · REAL-TIME · TRACEABLE

Quality decisions arrive too late

CURRENT WORKFLOW

- 1 Watch hundreds of changing sensor values
- 2 Detect drift after quality moves off target
- 3 Rely on experience to choose an adjustment
- 4 **Inspect output and absorb waste or rework**

GRADESENSE WORKFLOW

SENSE	Continuous process stream
PREDICT	Quality, risk, stabilization
EXPLAIN	Drivers, confidence, drift
RECOMMEND	Safe set-point changes

WHY IT MATTERS

- BEFORE DEFECTS
- ACTION, NOT ALERTS
- EXPLAINED DECISIONS
- CONTINUOUS LEARNING
- OPERATOR CONTROL

One governed path from sensor to action



TECHNOLOGY STACK

FRONTEND

React · TypeScript · Vite

BACKEND

FastAPI · Python · SQLAlchemy

ML

Scikit-learn · Pandas · NumPy

REAL TIME

REST APIs · WebSockets

DEPLOY

Docker · Render · PostgreSQL

LIVE MONITORING

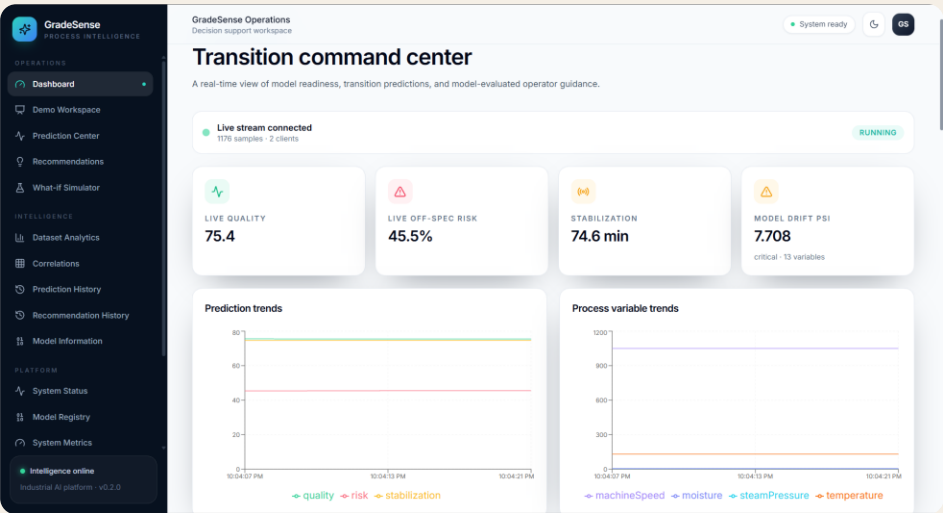
PREDICTIVE ANALYTICS

EXPLAINABLE AI

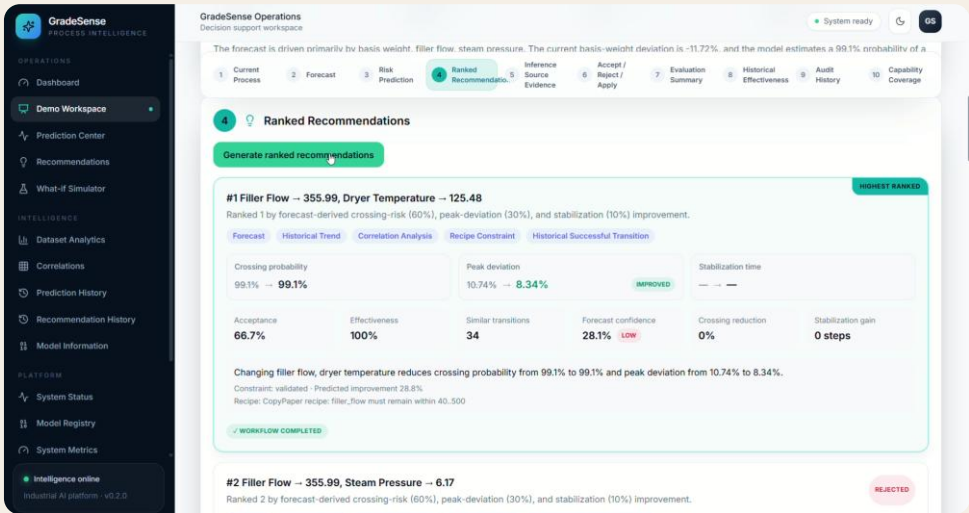
RANKED ACTIONS

AUDIT + HISTORY

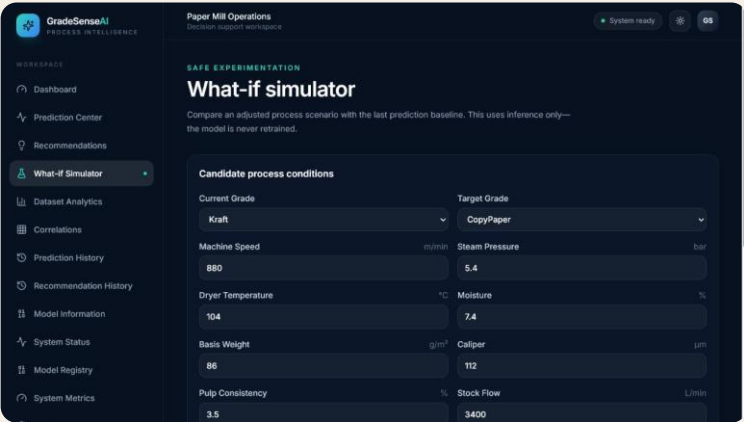
The closed loop is already working



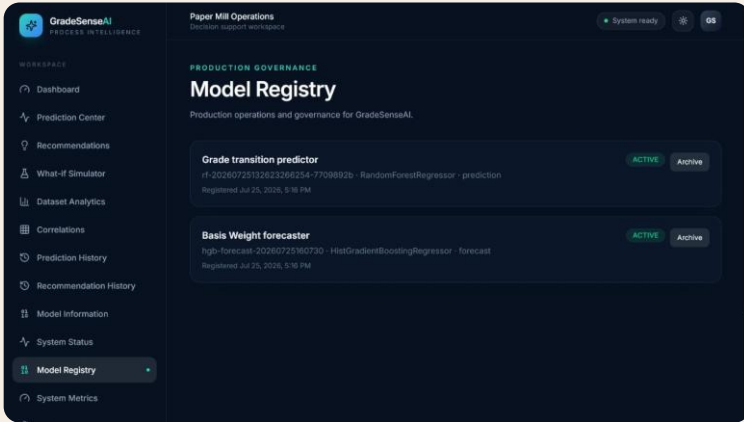
LIVE SIGNALS + EARLY RISK



EXPLAINED, RANKED ACTIONS



WHAT-IF BEFORE APPLYING



VERSIONED + AUDITABLE

Built to deploy - designed to earn trust

0.920

Crossing ROC-AUC

1.388

Trajectory MAE

0.962

Stabilization R²

2,544

Held-out windows

TECHNICALLY FEASIBLE

- Cloud deployed
- API-driven modules
- Real-time WebSockets
- Dockerized services
- Versioned model registry

BUSINESS VALUE

- Earlier intervention
- Faster operator decisions
- Lower rejection opportunity
- Improved process stability
- Higher trust through XAI

WHY GRADESENSE?

PREDICT



EXPLAIN



RECOMMEND



OPTIMIZE

One unified industrial decision-support platform.

Explore the working submission

GITHUB REPOSITORY

<https://github.com/bhavukm007/GradeSense>

LIVE DEPLOYMENT

<https://gradesense-4weh.onrender.com/>

DEMO VIDEO

<https://drive.google.com/file/d/1rCr3CzuCit0O029VzyxIhHoR1lv9E93E/view?usp=sharing>

TECHNOLOGY REFERENCES

React

FastAPI

Scikit-learn

Docker

Render

Pandas + NumPy

GRADESENSE

AI-powered intelligent process optimization platform